Year 10 · Chapter 1 · Financial mathematics: answers (teacher copy)

1.1 Percentages, fractions and decimals

(booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: 0.45
2. Ex 1b: 60%
3. Ex 1c: $\frac{1}{5}$
4. Ex 1d: 75%
5. Ex 2a: 0.1 × 80 = $8
6. Ex 2b: 0.25 × 36 = $9
7. Ex 2c: 0.15 × 240 = $36
8. Ex 2d: 21 ÷ 60 × 100 = 35%

Easy

1. 0.5
2. 0.25
3. 0.7
4. 0.09
5. 0.36
6. 1
7. 0.04
8. 0.85
9. 30%
10. 75%
11. 5%
12. 62%
13. 90%
14. 1%
15. 100%
16. 48%
17. $\frac{1}{2}$
18. $\frac{1}{4}$
19. $\frac{1}{10}$
20. $\frac{3}{4}$
21. $\frac{2}{5}$
22. $\frac{3}{10}$
23. $\frac{1}{100}$
24. $\frac{3}{5}$
25. 50%
26. 25%
27. 10%
28. 75%
29. 20%
30. 70%
31. 40%
32. 9%

Medium

1. $7
2. $23
3. $50
4. $7
5. $4
6. $9
7. $30
8. $45
9. $9
10. $42
11. $28
12. $10
13. $54
14. $15
15. $24.92
16. $10
17. 60%
18. 72%
19. 35%
20. 90%
21. 70%
22. 75%
23. 55%
24. 85%
25. 34 marks
26. $45
27. 280 students
28. 25%

1.2 Percentage increase and decrease

(booklet p. 5)

WE DO examples (teacher)

1. Ex 1a: 10% of $40 = $4 · $40 + $4 = $44
2. Ex 1b: 10% of $60 = $6 · $60 − $6 = $54
3. Ex 1c: 25% of $80 = $20 · $80 + $20 = $100
4. Ex 1d: 5% of $200 = $10 · $200 − $10 = $190
5. Ex 2a: 100% + 8% = 108% · 20 × 1.08 = $21.60
6. Ex 2b: 100% − 8% = 92% · 20 × 0.92 = $18.40
7. Ex 2c: 70 × 1.1 = $77
8. Ex 2d: GST = 110 ÷ 11 = $10

Easy

1. $22
2. $77
3. $33
4. $132
5. $99
6. $275
7. $550
8. $16.50
9. $45
10. $72
11. $18
12. $270
13. $63
14. $126
15. $360
16. $31.50
17. $60
18. $60
19. $72
20. $285
21. $125
22. $15
23. $420
24. $40
25. $66
26. $30
27. $180
28. $72
29. $21
30. $250
31. $8
32. $475

Medium

1. 1.15
2. 0.7
3. 1.2
4. 0.95
5. 1.5
6. 0.88
7. 1.03
8. 0.75
9. $92
10. $220
11. $1 084.16
12. $4 557
13. $47.70
14. $234
15. $1 236
16. $71.20
17. $44
18. $99
19. $275
20. $19.80
21. $2
22. $4
23. $9
24. $30
25. $5
26. $26 an hour
27. $15 300
28. 954 students

1.3 Profit, mark-up and discount (booklet p. 7)

WE DO examples (teacher)

1. Ex 1a: 20 − 12 = $8 profit
2. Ex 1b: 50 − 35 = $15 loss (it sold for less than it cost)
3. Ex 1c: 40 + 15 = $55
4. Ex 1d: 90 − 25 = $65
5. Ex 2a: mark-up = 0.25 × 40 = $10, so 40 + 10 = $50 (or 40 × 1.25)
6. Ex 2b: discount = 0.2 × 60 = $12, so 60 − 12 = $48 (or 60 × 0.8)
7. Ex 2c: 60 − 50 = $10 profit
8. Ex 2d: 10 ÷ 50 × 100 = 20%

Easy

1. $15
2. $15
3. $30
4. $15
5. $50
6. $7
7. $35
8. $60
9. $15 profit
10. $10 loss
11. $20 loss
12. $10 profit
13. $20 loss
14. $15 profit
15. $50 loss
16. $15 profit
17. $25
18. $50
19. $50
20. $100
21. $200
22. $20
23. $350
24. $80
25. $40
26. $60
27. $40
28. $90
29. $160
30. $90
31. $225
32. $24

Medium

1. $60
2. $88
3. $90
4. $240
5. $126
6. $45
7. $117
8. $73.60
9. $56
10. $60
11. $135
12. $36
13. $180
14. $72
15. $175
16. $74.80
17. 25%
18. 60%
19. 10%
20. 50%
21. 15%
22. 30%
23. 40%
24. 100%
25. selling price $11.40, profit $3.80
26. $4 200
27. $41.44
28. 45%

1.4 Wages and overtime (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: 20 × 8 = $160
2. Ex 1b: 600 × 2 = $1 200
3. Ex 1c: 600 × 52 = $31 200
4. Ex 1d: 52 000 ÷ 52 = $1 000
5. Ex 2a: 1.5 × 20 = $30 an hour
6. Ex 2b: 2 × 20 = $40 an hour
7. Ex 2c: 30 × 4 = $120

8. Ex 2d: normal pay 20 × 38 = $760 · overtime $120 (part c) · total $760 + $120 = $880

Easy

1. $100
2. $150
3. $180
4. $210
5. $760
6. $120
7. $1 000
8. $240
9. $800
10. $1 100
11. $1 200
12. $1 500
13. $1 800
14. $2 000
15. $1 640
16. $2 500
17. $5 200
18. $26 000
19. $52 000
20. $10 400
21. $31 200
22. $41 600
23. $15 600
24. $39 000
25. $1 000
26. $500
27. $2 000
28. $1 500
29. $1 250
30. $750
31. $1 100
32. $1 750

Medium

1. $30
2. $40
3. $45
4. $34
5. $36
6. $50
7. $24
8. $37.50
9. $120
10. $240
11. $96
12. $200
13. $144
14. $280
15. $132
16. $320
17. $950 + $75 = $1 025
18. $1 050 + $180 = $1 230
19. $800 + $180 = $980
20. $540 + $180 = $720
21. $473.40
22. $360
23. $912 + $180 = $1 092
24. day $508.80 + night $1 342.40 = $1 851.20

1.5 Commission, gross and net income

(booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: 0.1 × 800 = $80
2. Ex 1b: 0.05 × 2 000 = $100
3. Ex 1c: 300 + 150 = $450
4. Ex 1d: 250 + 40 + 75 = $365
5. Ex 2a: 0.035 × 4 600 = $161
6. Ex 2b: commission 0.05 × 3 000 = $150 · 400 + 150 = $550
7. Ex 2c: deductions 180 + 90 = $270 · 900 − 270 = $630
8. Ex 2d: gross 25 × 38 = $950 · net 950 − 190 = $760

Easy

1. $70
2. $50
3. $200
4. $30
5. $150
6. $400
7. $65
8. $1 200
9. $100
10. $90
11. $30
12. $200
13. $95
14. $20
15. $150
16. $500
17. $580
18. $450
19. $750
20. $650
21. $840
22. $285
23. $895
24. $575
25. $600
26. $700
27. $740
28. $550

Medium

1. $37.50
2. $200
3. $360
4. $44.10
5. $57.60
6. $91
7. $70.50
8. $75.80
9. $500
10. $450
11. $530
12. $554
13. $440
14. $320
15. $630
16. $410
17. $840
18. $560
19. $689.50
20. $645.60
21. 0.045 × 790 = $35.55
22. commission 0.02 × 45 000 = $900 · 500 + 900 = $1 400
23. deductions 230 + 114 + 15 = $359 · 1 200 − 359 = $841
24. gross wage $1 040.04, net pay $874.95

1.6 Simple interest and loans (booklet p. 13)

WE DO examples (teacher)

1. Ex 1a: 5 ÷ 100 = 0.05
2. Ex 1b: I = 1 000 × 0.05 × 2 = $100
3. Ex 1c: I = 400 × 0.1 × 3 = $120
4. Ex 1d: I = 500 × 0.1 × 2 = $100 · total 500 + 100 = $600
5. Ex 2a: I = 2 500 × 0.045 × 3 = $337.50
6. Ex 2b: 2 × 12 = 24 repayments · 150 × 24 = $3 600
7. Ex 2c: 3 600 − 3 000 = $600
8. Ex 2d: deposit 0.1 × 2 000 = $200 · total 200 + 90 × 24 = $2 360 · interest 2 360 − 2 000 = $360

Easy

1. 0.05
2. 0.1
3. 0.03
4. 0.2
5. $20
6. $50
7. $5
8. $30
9. $20
10. $60
11. $80
12. $80
13. $100
14. $150
15. $80
16. $40
17. $300
18. $140
19. $120
20. $180
21. $600
22. $1 150
23. $300
24. $480
25. $2 120
26. $1 040
27. $140
28. $3 300

Medium

1. $131.25
2. $1 600
3. $64.50
4. $20
5. $250
6. $360
7. $600
8. $900
9. $12 000
10. $9 000
11. $3 960
12. $2 292
13. $400
14. $200
15. $1 400
16. $820
17. $120
18. total repaid $12 000, interest $7 000
19. deposit $120, total paid $1 320
20. deposit $1 050, total repaid $13 050, interest $6 050

1.7 Compound interest (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: 0.1 × 1 000 = $100
2. Ex 1b: 1 000 + 100 = $1 100
3. Ex 1c: 10% of 1 100 = $110 · 1 100 + 110 = $1 210
4. Ex 1d: I = 1 000 × 0.1 × 2 = $200, balance $1 200 · compound gives $10 more
5. Ex 2a: R = 0.06 · 1 + 0.06 = 1.06
6. Ex 2b: A = 1 000 × 1.1^3 = $1 331
7. Ex 2c: A = 2 500 × 1.06^4 = $3 156.19
8. Ex 2d: 3 156.19 − 2 500 = $656.19

Easy

1. $10
2. $200
3. $100
4. $100
5. $330
6. $1 100
7. $420
8. $600
9. $250
10. $770
11. $3 150
12. $720
13. $121
14. $242
15. $605
16. $1 210
17. $363
18. $2 420
19. $484
20. $726
21. interest $200, $220, $242 · balance $2 200, $2 420, $2 662 · total interest $662
22. simple $600, $700, $800 · compound $600, $720, $864 · $64 more

Medium

1. 1.1
2. 1.03
3. 1.2
4. 1.08
5. 1.12
6. 1.04
7. 1.025
8. 1.045
9. $1 210
10. $665.50
11. $2 315.25
12. $1 152

13. $1 622.40
14. $675.31
15. $4 307.56
16. $3 345.56
17. $102.50
18. $630
19. $291.20
20. $185.45
21. $832
22. $149.84
23. $196.86
24. $1 040.81
25. value $5 624.32, interest $624.32
26. A = $5 770.24, interest $1 170.24
27. simple $300, compound $315.25, $15.25 more
28. A = $6 022.53, interest $1 022.53